Inorganic Cumulative Exam
March 4 1995

The electronic spectra of transition metal complexes were of key importance in the developments of bonding theories for this class of compounds. Spectral measurements continue to be of the special interest among inorganic chemists and can provide valuable information about the complexes and potential reactivity. The following terms often appear in this context.

1. Russell Saunders terms symbols, what do they tell us? Give an example.

2. What is a Tanabe-Sugano Diagram? How are the axes defined and how is it used?

3. What does a Racah parameter describe?

4. Some transitions are said to be LaPorte forbidden. What does this mean?

5. Some transitions are said to be parity forbidden. What does this mean?

6. Some transitions are said to be spin forbidden. What does this mean?

7. Give an example of a complex that suffers from the feature described in 4 and 6.

8. Using an appropriate MO diagram, illustrate a charge transfer transition.

9. How are ionization energies related to the charge transfer energies? Use an example to describe the relation.

10. What is the difference between fluorescence, luminescence and phosphorescence?